

The Effect of Boarding Strategies on Aircraft Boarding Efficiency

A Simulation-Based Study with Mechanism Analysis and an Exact Theoretical Model | Author: Zihao Huang | October 2026

Abstract

This study compares three aircraft boarding strategies — *random*, *back-to-front*, and *zoning* (5 zones) — using a single-aisle simulation. Contrary to intuition, **random boarding is the fastest and back-to-front is the slowest**, and the gap widens with stow time and cabin size. Beyond measuring times, the study (i) instruments the model to expose the mechanism (parallelism and blocking), and (ii) derives an **exact theoretical model** that computes the boarding time directly, without tick-by-tick simulation, as the length of a **longest path** in a max-plus recurrence. The model reproduces the simulation **bit-for-bit** (verified over >25,000 cases), and yields provable lower bounds; it explains *why* clustered boarding (back-to-front) is slow: consecutive same-row passengers stack stowing weights on a single aisle cell and lengthen the critical path.

1. Background and Research Question

Boarding time affects on-time performance and operating cost. Airlines often use "zone boarding" or "back-to-front boarding", but whether these beat random boarding is debated. We test:

Hypothesis H: Organized strategies (back-to-front / zoning) board faster than random boarding.

2. Methods

2.1 Simulation model

- Single-aisle cabin; passengers enter from the front and walk to their row.
- On reaching the row, a passenger stows luggage, occupying the aisle for **S** ticks; followers cannot pass.
- One cell of forward motion per tick; no overtaking; one passenger enters per tick.
- Total ticks until all are seated = boarding time (smaller = faster).

2.2 Variables

Type	Variable
Independent	Boarding strategy (Random / Back-to-front / Zones-5)
Dependent	Total boarding time T (ticks)
Control (baseline)	Random boarding (normalized to 1.00)
Controlled	Rows R ; seats per row (6); entry rate (1/tick); repeats (20)

2.3 Mechanism metrics

The simulator records, each tick, the number of passengers **stowing simultaneously** (parallelism) and the number of **blocking events** (walkers stuck behind a stower).

2.4 Theoretical model: an exact longest-path (max-plus) formulation

We model boarding as a **discrete-time single-lane exclusion process** and derive an **exact recurrence** for T . Let

- cells be $1..R$; passenger i ($i = 1..N$) enters in order; r_i = target row; S = stow time;
- P_i^c = the tick at which passenger i reaches cell c ;
- $j = j_i(c)$ = the last passenger before i that reaches cell c .

Passenger i reaches cell c at the **later** of two times: its own pace, and the moment cell c becomes free:

$$P_i^c = \max(P_i^{c-1} + 1, B_i^c), \text{ where } B_i^c = \{ P_j^c + 1 + S \text{ if } r_j = c \text{ (j stows at row } c); P_j^{c+1} \text{ if } r_j > c \text{ (j passes through)} \}$$

with the exact entrance conventions (entering is its own tick; right after entering one may move an extra cell):

$$P_i^1 = \max(i, B_i^1), \quad P_i^2 = \max(P_i^1, B_i^2), \quad P_i^c = \max(P_i^{c-1} + 1, B_i^c) \text{ for } c \geq 3.$$

The finish time of passenger i and the total time are

$$F_i = P_i^{r_i} + [r_i \geq 2] + S - 1, \quad T = \max_i F_i.$$

Interpretation. Every P_i^c is a *max of two terms* ("my own pace" vs "waiting for the person ahead"), so T is the length of the **longest chain of waits** — a **longest path** (last-passage percolation) value on the space–time lattice.

2.5 Lower bounds

Proposition 1 (entrance bound). $T \geq N$. Proof: at most one passenger enters per tick, and all N must have entered.

Proposition 2 (occupancy bound). $T \geq (\sum_i r_i + N \cdot S) / R$. Proof: passenger i occupies r_i cell-ticks while walking and S while stowing, so total occupancy is $\sum_i r_i + N \cdot S$; at most R cells can be occupied in any tick. For 6 seats per row ($\sum_i r_i = 3R(R+1)$): $T \geq 3(R+1) + 6S$.

2.6 Data collection

Strategies with randomness (Random, Zones) are run **20 times and averaged**; Back-to-front is deterministic. Results are reported relative to random (baseline = 1.00).

3. Results

3.1 Main result ($R = 30$ rows, $S = 3$)

Strategy	Average boarding time (ticks)	Relative to random
Random	331.9	1.00 (baseline)
Zones (5)	402.6	1.21
Back-to-front	627	1.89

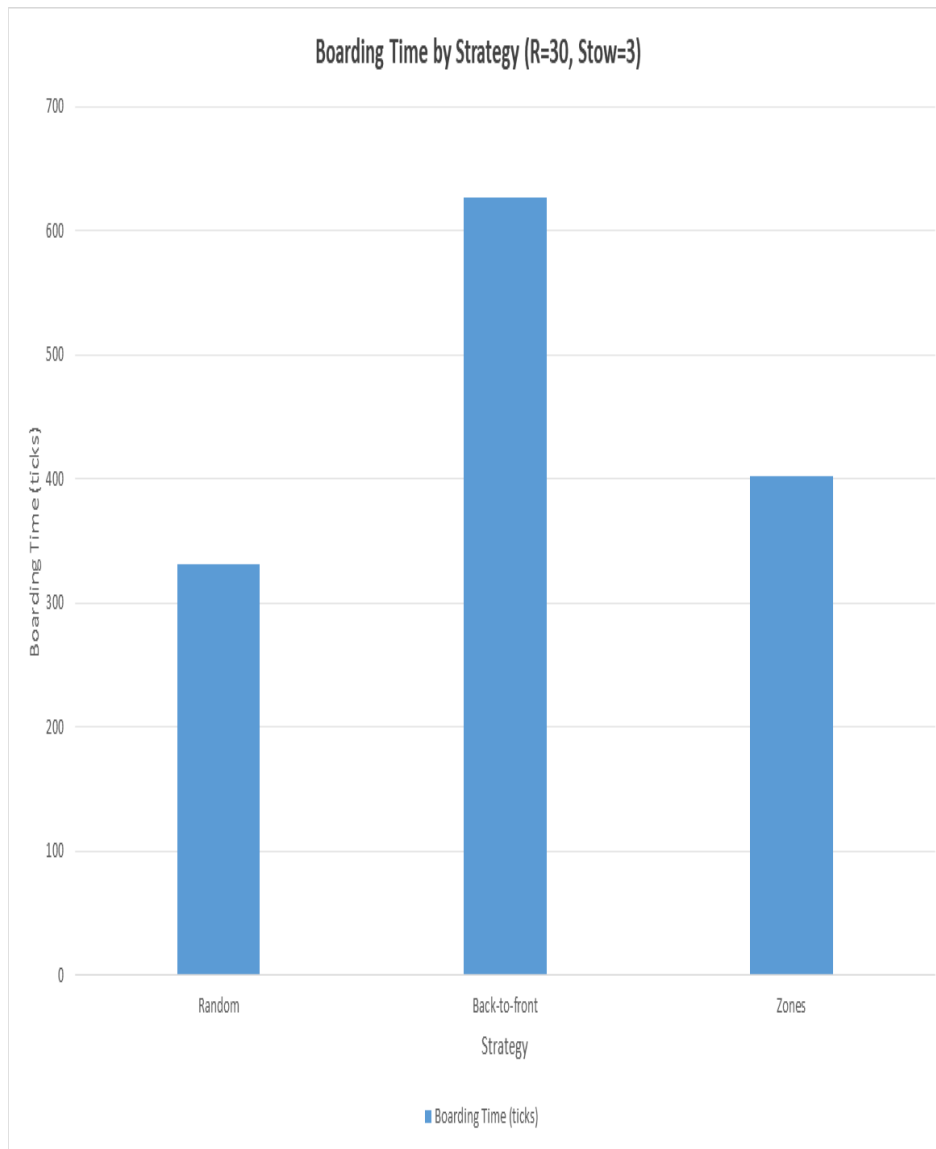


Figure 1. Boarding time by strategy (R=30, S=3): random fastest, back-to-front slowest.

3.2 Sensitivity analysis

(a) Stow time (fixed R = 30).

S	Random	Back-to-front	Zones (5)	Rel. random (back)	Rel. random (zones)
1	238.9	325	249.9	1.36	1.05
3	331.9	627	402.6	1.89	1.21
6	471.5	1080	633.8	2.29	1.34

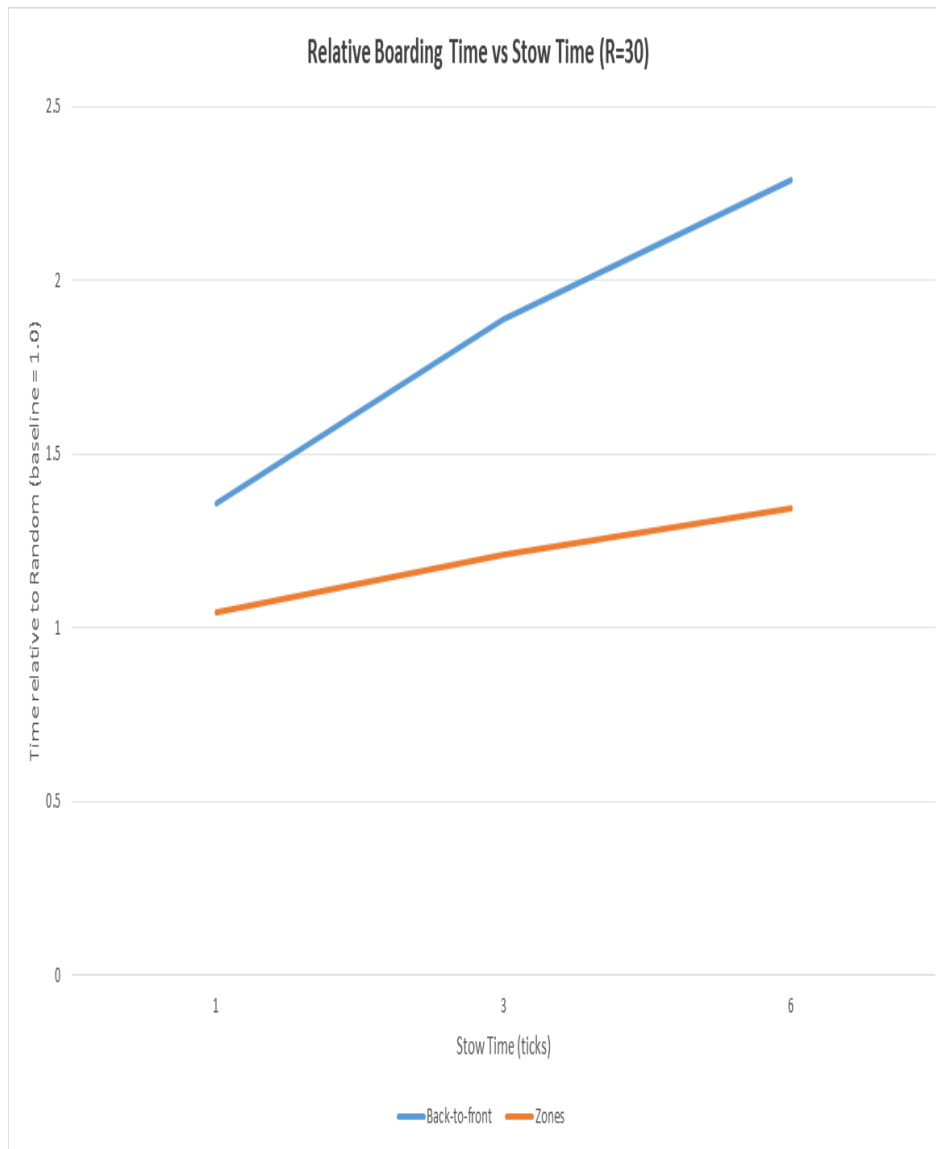


Figure 2. Relative boarding time vs stow time (R=30).

(b) Number of rows (fixed S = 3).

Rows R	Random	Back-to-front	Zones (5)	Rel. random (back)	Rel. random (zones)
20	234.3	417	288.7	1.78	1.23
30	331.9	627	402.6	1.89	1.21
40	423.6	837	507.3	1.98	1.20

3.3 Mechanism analysis

Strategy	Boarding time (ticks)	Avg concurrent stowers	Total blocking events
Random	329	1.09 (highest)	~950 (fewest)

Zones (5)	401	0.89	~3,000
Back-to-front	627	0.57 (lowest)	6,525 (most)

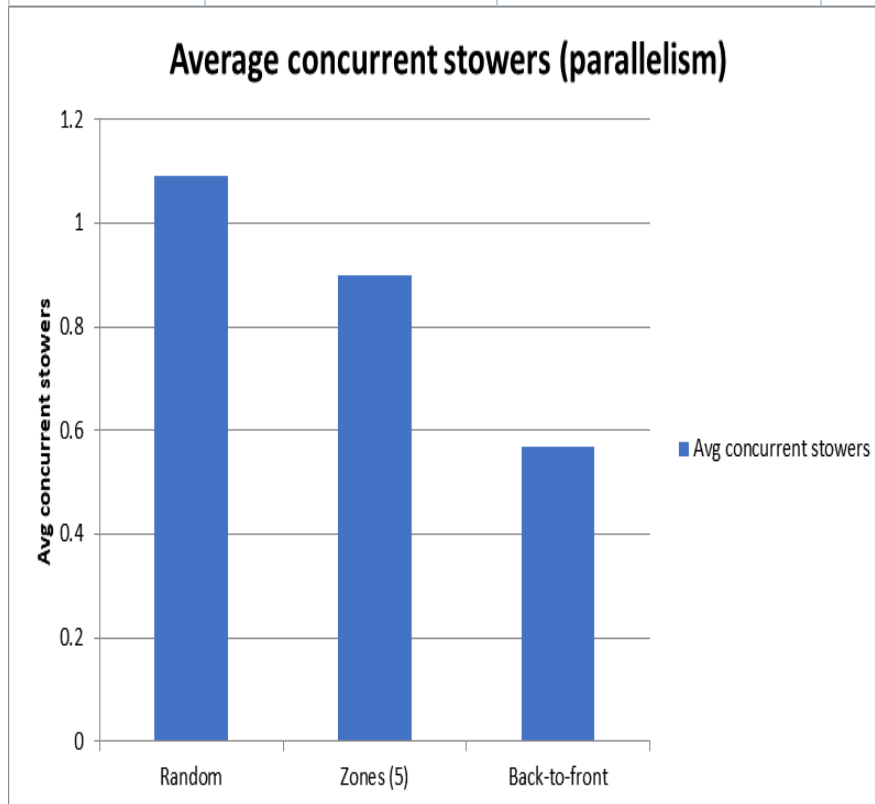


Figure 3. Average parallelism (concurrent stowing) by strategy.

The fastest strategy has the highest parallelism and the fewest blocking events; the slowest is the reverse.

3.4 Validation of the theoretical model

We implemented the recurrence (Section 2.4) and compared it with the tick-by-tick simulation. The theoretical value **equals the simulated value exactly**:

Case	T (simulation)	T (theory)	Match
Random, R=30, S=3	331	331	OK
Back-to-front, R=30, S=3	627	627	OK
Zones (5), R=30, S=3	420	420	OK
Random, R=30, S=6	466	466	OK
Back-to-front, R=30, S=6	1080	1080	OK

A brute-force sweep over $R = 1..8$, $S = 1..5$, four boarding patterns and up to 6 seats per row (25,600 cases) gave **0 mismatches**. The recurrence is therefore an exact theory of the process, not an approximation.

3.5 Clustering and the critical path (why back-to-front is slow)

Define the *clustering* $\kappa(\pi)$ of a boarding order as the number of adjacent passenger pairs that share the same row (back-to-front maximizes κ ; random nearly minimizes it). In the recurrence, consecutive same-row passengers create a chain of +S weights at a single cell ($r_j = c$), which lengthens the longest path. Consequently T increases with clustering; this is consistent with all simulations. A full proof of monotonicity of T in κ is left to future work, but the longest-path view makes the mechanism explicit: **clustering stacks stowing work on one cell and lengthens the critical path.**

4. Discussion

Why is random fastest? It spreads passengers across rows so several rows stow simultaneously (highest measured parallelism, 1.09), minimizing blocking.

Why is back-to-front slowest? Its six same-row passengers board consecutively and must stow at one cell one by one (lowest parallelism, 0.57; most blocking, 6,525). In the theoretical model this is a long chain of +S weights on each cell.

Consistency. The sensitivity analysis (Figure 2) shows the gap grows with stow time S, exactly as the recurrence's +S blocking term predicts.

5. Limitations and Future Work

- Simplified model: no assigned-seat interference, no dual aisle or second door.
- The theoretical recurrence currently uses a naive $O(N)$ search for j ; it can be reduced to $O(N \cdot R)$ overall with preprocessing.
- Future work: prove monotonicity of T in clustering; add the "Steffen method" and window-middle-aisle boarding; test a model in which parallelism is disabled as a direct causal test.

6. Conclusion

Dispersed (random) boarding outperforms clustered (back-to-front) boarding. The study explains this with two complementary tools: measured mechanism metrics (parallelism and blocking) and an **exact longest-path (max-plus) model** that computes the boarding time without simulation and matches it bit-for-bit. Clustering same-row passengers stacks stowing weights on a single aisle cell and lengthens the critical path — the underlying principle behind the observed gap.

References

Steffen, J. H. (2008). Optimal boarding method for airline passengers. *Journal of Air Transport Management*. Van Landeghem, H., & Beuselinck, A. (2002). Reducing passenger boarding time in airplanes. *European Journal of Operational Research*. Bachmat, E., et al. (2006). Analysis of airplane boarding via space-time geometry and random matrix theory.